

Efficient Analysis and Minimization of Glitches using Threshold Swapped Combinational Clock Gating

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Abstract

Nowadays design of electronic circuits to meet requirements like high speed, better functionality and low power is a tedious task. Power dissipation in any circuit comprises mainly two components: static power and dynamic power. The dynamic power heavily depends on the switching activity of the circuit. During the transistor switching transitions some spikes may occur, which are called as glitches. These glitches contribute 20–70% of the dynamic power and 20% of the total power consumption. Glitches will occur at a particular candidate gate if the delay of the gate is less than the differential delay of their paths at the inputs. Numerous techniques have been proposed so far for suppression of glitches. But all these techniques target for either low power or speed. Here we propose a novel technique based on effective utilization of fast, slow and typical library cells and clock gating technology by targeting for zero dynamic power. In this paper, we have thoroughly discussed about the effects occurring due to the proposed technique on area, delay and power of the ISCAS 85 bench mark circuits using Cadence RTL compiler.

Keywords: Low power design, glitches, clock gating, multi threshold, zero leakage, zero dynamic

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INTRODUCTION

Low power design of digital circuits is a crucial issue in electronics industry. Power dissipation in digital circuits mainly consists of functional switching, glitches, short circuit and leakage power. Functional transitions are needed to perform the tasks. Hazards or glitches are spurious pulses which occur before steady state is reached. During switching of the transistors short circuit power arises. Due to faulty transistors leakage power will occur which is very small component [1]. Glitches are classified into two types: generated glitches and propagating glitches. Glitches are generated in the circuit due to difference in arrival times of the input signals from different paths of the gate. Generated glitches occur during the functional transitions, whereas propagating glitches occur due to input signals [2]. Hashimoto *et al.* proposed a heuristic glitch reduction technique based on sensitivity of the gate in any logic circuit. Statistical approaches are adopted in the estimation of glitches.

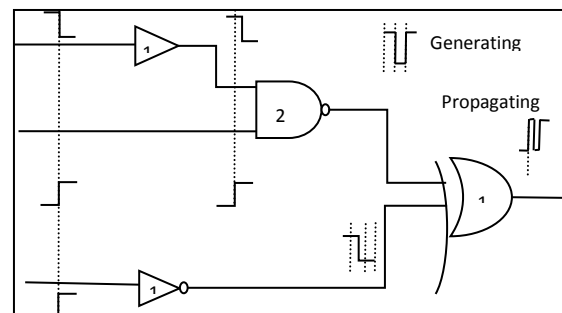


Fig. 1: Glitches Generation and Propagation.

As per the sensitiveness of a particular gate for the given input stream, that gate may be down sized or upsized in order to block generated glitches from predecessor gate [3]. This method provides best way for modeling of both generated glitches and propagating glitches. Hill climbing process like this involves complex calculations and reported increase in area. Kim *et al.* proposed another technique for glitch power reduction by making use of path balancing approach. In order to reduce glitches as well as load capacitance this algorithm works well. Buffers

are inserted in non-sized paths using inductive logic programming (ILP) [4]. Path balancing achieves equal arrival times at the input of the gate for all possible paths. Benini *et al.* developed a technique to minimize spurious spikes in logic circuit by replacing existing gates with functionally equivalent gates F-Gates. In this technique, n-type transistor is connected in series with n-pull down network controlled by control signal. When control signal is zero the output node is disconnected from ground, in which discharge path is not established from one to zero. Hence one to zero transition is suppressed. Control signal is generated based on the various parameters like gate's arrival time, required time and slack. Gate freezing technique reduces glitch power only due to the suppression of one to zero transitions. Glitch power is reduced without much area head like sizing and buffer insertion algorithm. But, zero to one transition is not avoidable and overall delay of the circuit might boost up due to addition of control signal to the circuitry [5].

Lee *et al.* proposed another technique for hazard power minimization to establish zero difference delay paths at the input of the gate terminals and reported good results towards less glitch power. Sizing of gates, insertion of buffers, adding control signal circuitry overall delay may increase [2]. Slimani *et al.* made use of threshold voltage for glitch power reduction. In MOSFET, threshold voltage is expressed by:

$$V_{th} = V_{th0} + \gamma \sqrt{2\phi_F + V_{sb}} - \sqrt{2\phi_F}$$

Where V_{th0} =threshold voltage at zero substrate, ϕ_F =Fermi potential, γ =body bias coefficient, V_{sb} =Source bulk voltage.

All these methods work on non-critical paths to compensate differential path delays of the candidate gate. Insertion of buffers, varying size of the transistors leads to area overhead in digital circuits and these may suffer from much computation time [6]. In this paper, power optimization is achieved with very little overhead of delay and area. We made use of threshold voltage adjustment and clock gating procedures for achieving our goal. In later sections of this paper, we will discuss the effect of threshold voltage on delay and power with a mathematical model, clock gating with multi threshold technique with its effects and various performance metrics of proposed algorithm and their comparison with the existing algorithms will be provided.

EFFECT OF THRESHOLD VOLTAGE ON DELAY AND POWER

The main cause of glitches is different arrival times at the input of the gate terminals. These hazards can be suppressed by proper adjustment of path delays and gate delays. Figure 2 shows the generation and suppression of glitches due to adjustment of gate delays and this method is termed to be glitch filtering [7].

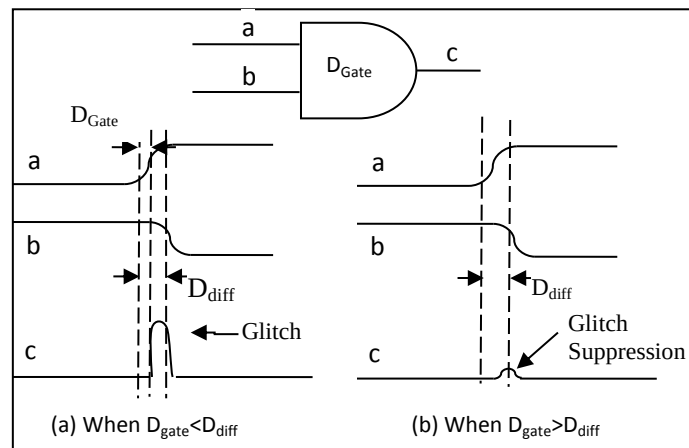


Fig. 2: Suppression of Spikes with Delay Adjustment.

Spikes or glitches at the output node of the gate are at observable level whenever the gate switches more than once before attaining

steady state for inputs as shown in Figure 2. If the candidate gate's delay D_{Gate} is higher than the value of differential path delay the

spurious pulses are suppressed and the gate is said to be glitch free one. Due to increasing of gate delay to suppress glitches overall delay overhead might increase [6]. Over recent years numerous methods have been developed to solve the various aspects of power problem and to meet ever seen power requirements. These include clock gating, multi threshold (Multi Vt) transistors, multi supply voltage, substrate biasing, dynamic voltage and frequency scaling and power shut off [8]. For the targeted timing delay, the technologies less than 90 nm design for low power dissipation have plenty of limitations. A design is to start with typical cells, once the timing target is met roughly non critical path, cells are mapped with slow cells, and critical path cells with fast cells due to their speed. Real complication lies with mapping of these cells. Timing and power minimization are like head and tail of a coin. In this paper, we present the effect of threshold voltage on performance metrics like delay and power by considering four stages of CMOS inverter as shown in Figure 3 with three cases like high V_{th} ($V_{th} > V_{DD}/2$)-Slow cells, Medium V_{th} ($V_{th} < V_{DD}/2$)-Typical cells and Low V_{th} ($V_{th} < V_{DD}/2$)-Fast Cells.

Case 1: $V_{TH} > V_{DD}/2$

Initially consider nMOS M_2 is in OFF state, pMOS M_1 in ON state and the output node of

first stage V_{C1} is charged to V_{DD} . During positive going step at the input of the first stage of inverter, V_{C1} starts discharging through nMOS M_2 as shown in Figure 4.

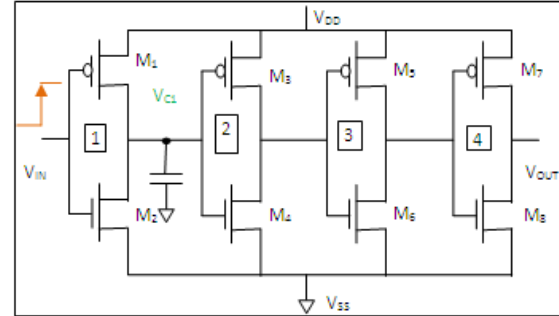


Fig. 3: Cascading of CMOS Inverters.

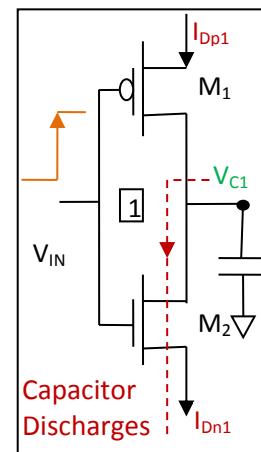


Fig. 4: Capacitor Discharging Path.

During this transition, M_2 is in saturation region until V_{C1} discharges up to $V_{GS} - V_{TH}$. The expression for current in saturation region [9] is given by:

$$i_{Dn1} = \mu_n C_{ox} \frac{W_n}{L_n} \frac{(V_{GS} - V_{TH})^2}{2} = k_{1n} (\text{Constant}) \quad (1)$$

By integrating Eq. (1) and keeping boundary condition at $t=0$, $V_{C1}=V_{DD}$, V_{C1} can be expressed as:

$$V_{C1} = -\frac{k_{1n}t}{C_L} + V_{DD}$$

When $t=T_1$ the capacitor reaches to a value of $(V_{GS} - V_{TH})$

$$V_{C1} = (V_{GS} - V_{TH}) = -\frac{k_{1n}T_1}{C_L} + V_{DD} \quad (2)$$

Where T_1 is the time required by the capacitor to discharge upto $V_{GS} - V_{TH}$. When V_{C1} discharges below $(V_{GS} - V_{TH})$, nMOS M_2 will enter into triode region. The current flowing through the capacitor by neglecting second term [9] can be expressed as:

$$i_{Dn1} = \mu_n C_{ox} \frac{W_n}{L_n} (V_{GS} - V_{TH}) V_{C1} = K_{2n} V_{C1} \quad (3)$$

Voltage across capacitor during this discharge time gives linear differential equation as:

$$\frac{dV_{C1}}{dt} + \frac{K_{2n}V_{C1}}{C_L} = 0 \quad (4)$$

The solution for Eq. (4) and applying boundary condition at $t=T_1$, $V_{C1}=V_{GS}-V_{TH}$ is:

$$V_{C1} = (V_{GS} - V_{TH}) e^{-\frac{t}{\tau_2}} \quad (5)$$

If T is the time required by the capacitor to discharge completely approximately $0.1 V_{DD}$ and τ_2 is the time constant equals to C_L/K_{2n} . Hence Eq. (5) can be written as:

$$V_{C1} = 0.1 V_{DD} = (V_{GS} - V_{TH})e^{-\frac{T}{\tau_2}} \quad (6)$$

Total time for capacitor C_L to get discharge completely ($0.1 V_{DD}$) is T (during T_1 nMOS is in saturation and during $T-T_1$ nMOS is in triode region). Similarly if the input transition is from one to zero, pMOS goes to ON state and nMOS driven to OFF state. Capacitor C_L charges to V_{DD} . V_{C1} charges with constant current till its voltage reaches upto $|V_{THP}|$, after this pMOS enters into triode region.

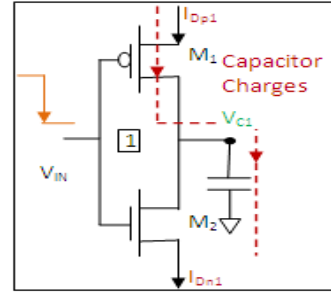


Fig. 5: Capacitor Charging Path.

When $V_{C1} < |V_{THP}|$

$$i_{DP1} = \mu_p C_{ox} \frac{W_p (V_{SG} - |V_{THP}|)^2}{L_p} = K_{1p} (\text{Constant})$$

Where; μ_p the mobility of pMOS carriers, C_{ox} the oxide layer thickness, W_p and L_p represent width and length of pMOS transistor, $V_{SG} = (V_{DD} - V_G)$ is gate to source voltage and V_{THP} is threshold voltage of pMOS transistor. Voltage across capacitor is given as:

$$V_{C1} = \frac{K_{1p} t}{C_L} \quad (7)$$

If V_{C1} reaches $|V_{THP}|$, pMOS transistor M_1 enters into triode region, then current expression becomes:

$$i_{DP1} = \mu_p C_{ox} \frac{W_p}{L_p} (V_{SG} - V_{THP}) V_{SD} = K_{2p} (\text{Constant})$$

$$i_{DP1} = \mu_p C_{ox} \frac{W_p}{L_p} (V_{DD} - V_G - V_{THP})(V_{DD} - V_{C1}) = K_{2p} (\text{Constant}) \quad (8)$$

To obtain voltage across capacitor, integrate Eq. (8) with respect to time and using boundary condition at $t=T_1$, $V_{C1}=V_{THP}$.

$$V_{C1} = V_{DD} + (V_{THP} - V_{DD})e^{-\frac{t-T_1}{\tau_4}} \quad (9)$$

Where $\tau_4 = C_L / K_{2p}$ and T_1 is the time required to charge upto V_{THP} . Eq. (2) and (6) together give the expressions for output node for 0 to 1 transition at the input and Eq. (7) and (9) give the output node expressions for the input 1 to 0. These expressions will drive next stages of the inverters.

Case 2: $V_{TH} < V_{DD}/2$

During this stage $V_{TH} < V_G < (V_{DD} - |V_{THP}|)$, C_L starts charging from V_{DD} and some current will be draining to ground as nMOS is also in ON state which is short circuit power for the duration of T' . T' is the time required by the gate voltage to change from $(V_{DD} - |V_{THP}|)$ to V_{TH} . During T' pMOS is in saturation region and nMOS in triode region.

$$\begin{aligned} I_{DP2} &= \mu_p C_{ox} \frac{W_p (V_{SG} - V_{THP})^2}{L_p} \\ &= \mu_p C_{ox} \frac{W_p (V_{DD} - V_G - V_{THP})^2}{L_p} = K_{1p} \end{aligned}$$

$$\begin{aligned} I_{DN2} &= \mu_n C_{ox} \frac{W_n}{L_n} (V_{GS} - V_{THn}) V_{C2} \\ &= K_{2n} V_{C2}, \end{aligned}$$

This equation is valid when $V_{C2} < V_G - V_{TH}$.

$$V_{C2} = \frac{1}{C_L} \int (I_{DP2} - I_{DN2}) dt \quad (10)$$

By solving Eq. (10) becomes:

$$V_{C2} = \frac{K_{1p}}{K_{2n}} \left(1 - e^{-\frac{t}{\tau_5}} \right) \quad (11)$$

Where;

$$\tau_5 = \frac{C_L}{K_{2n}}$$

$$K_{1p} = \mu_p C_{ox} \frac{W_p}{L_p} (V_{SG} - V_{THP})^2 0.5$$

$$K_{2n} = \mu_n C_{ox} \frac{W_n}{L_n} (V_{GS} - V_{TH})$$

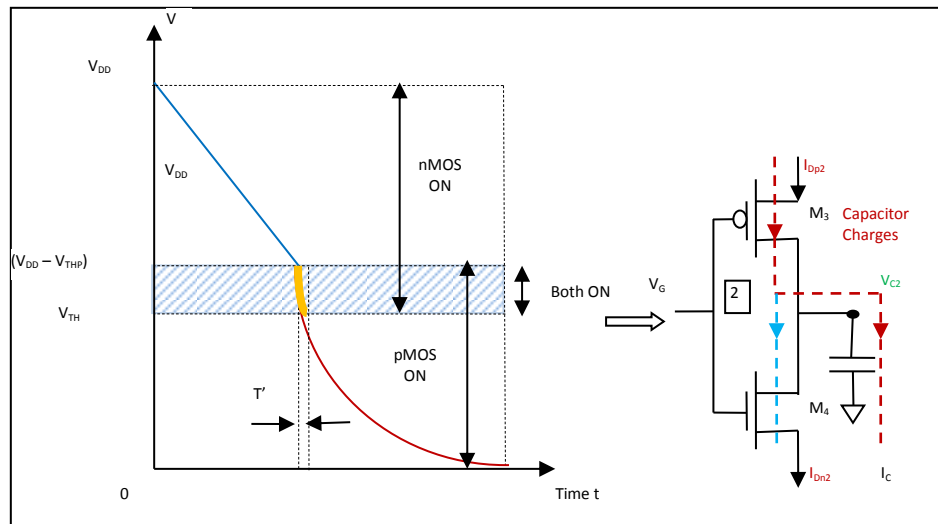
$$K_{2n} = \mu_n C_{ox} \frac{W_n}{L_n} (V_G - V_{TH})$$


Fig. 6: *Discharging/Short Circuit Path during T'.*

$$V_{G2} = |V_{THP}| = \frac{K_{1p}}{K_{2n}} \left(1 - e^{-\frac{t}{\tau_5}} \right) \quad (15)$$

When $T_1 < t < T'$ (pMOS in triode region and nMOS is in ON state)

$$V_{C2} = \frac{V_{DD}K_{P'}}{K_{N'}} + \left[V_{THP} - \frac{V_{DD}K_{P'}}{K_{N'}} \right] e^{-\frac{t-T_1}{\tau_8}} \quad (16)$$

By plotting Eqs. (9), (12), (13), (14), (16) and (17) using MATLAB, the behavior of MOS device for various threshold voltages is as shown in below.

Page 33

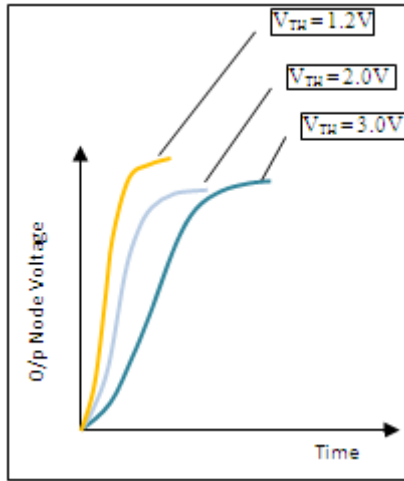


Fig. 7: Delay Calculations for Different Threshold Voltages.

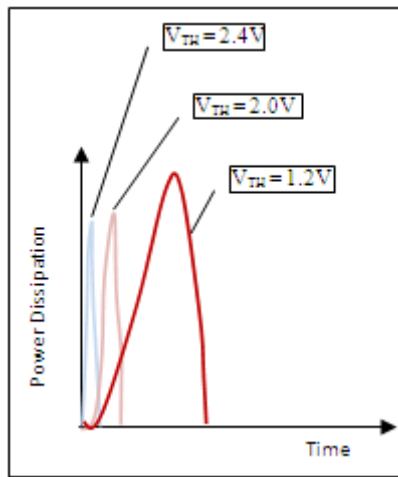
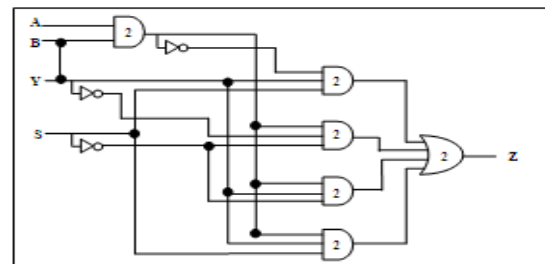


Fig. 8: Power Calculations for Different Threshold Voltages.

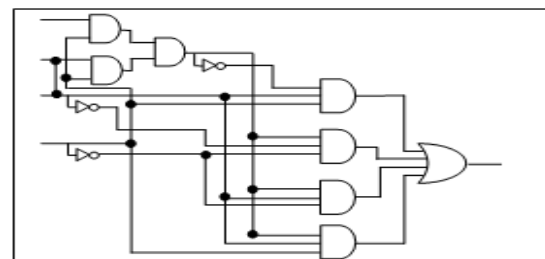
Figure 7 clearly describes that the time required for charging output node capacitor decreases with decrease in threshold voltage. Figure 8 proves that power of the circuit increases with decrease in threshold voltage. These two statements are contradiction for multi objective design closure. From this mathematical analysis we can say that high V_t (slow cells) cells provide more delay and less short circuit power, Medium V_t cells (typical cells) provide moderate performance and Low V_t cells (fast cells) give less delay with more leakage power. For multi objective design closure, it is contradiction to make use of cells from any one library. In this paper, a heuristic technique is proposed for optimum power, delay and area.

THRESHOLD SWAPPED CLOCK GATING

Power consumption of an IC can be controlled at maximum extent by minimizing switch ON and OFF timings of a transistor [10] and also can be minimized by selectively assigning common low power supply to the highly active regions [11]. Threshold swapping of cells in critical paths greatly, also effects system power at large extent [12]. Gating technique mainly works on the principle that reduction of effective frequency or toggling frequency of any circuit during wasted operations [13]. Clock gating first defined in synthesis stage and then optimized in implementation stage. This technique would generally occur in two stages cloning and de cloning [14]. In SOC design, clock gating technique is successfully implemented and achieved best results [15]. Along with clock gating, if V_{th} scaled properly glitches can be suppressed [16]. Clock gating reduces dynamic power produced by the clock distribution network in digital circuits [17]. Clock gating techniques supported by the most of the commercial EDA tools for logic synthesis can be applicable and are effective at different levels of application [18]. Various approaches in clock gating at register level and system level are proposed in Refs. [19, 20]. Figure 9 clearly explains the procedure of clock gating and shows how clock gating can be achieved.



(a) Before Optimization.



(b) After Optimization.

Fig. 9: Modified Architecture using Clock Gating.

As shown in Figure 9(a), the multiplier whose inputs change for every cycle and output fed to multiplexer conditionally. In optimized version, inputs are prevented from rippling through multiplier if it is not selected by multiplexer. By using this technique, we can achieve low power with little area overhead. This architecture is further optimized by incorporating multi threshold cells. In this paper, we present results obtained for combination of clock gating and multi threshold techniques by targeting device for zero leakage power and zero dynamic power.

RESULTS AND DISCUSSIONS

In this paper, we present the results obtained for multi objectives like area, delay and power. The proposed technique is investigated exhaustive lyon ISCAS 85 benchmark circuits using Cadence RTL Compiler. Obtained results are then compared with the results obtained for multi threshold technique, gate

sizing and clock gating as shown in Tables 1 to 8.

GS and B: Gate sizing and buffer insertion.

CG: Clock gating.

MV_t: Multiple threshold voltage.

MV_t+CG: Multiple threshold with clock gating.

Zero Leakage: Swapping of cells targeting for zero leakage.

Zero Dynamic: Swapping of cells targeting for zero dynamic.

Table 1: Performance Comparison for Area Design Closure for Existing Techniques.

Ckt	GS and B	CG	MV _t
C432	2005.819	1051.142	1057.795
C499	3778.79	3695.63	3535.963
C880	3708	2564.654	2514.758
C1355	5202.49	3718.915	3506.026

Table 2: Performance Comparison for Area Design Closure for Proposed Techniques.

Ckt	MV _t +CG	MV _t -Zero Leakage		MV _t -Zero Dynamic	
		w/o CG	with CG	w/o CG	with CG
C432	1047.816	1044.49	1044.49	1057.795	1047.816
C499	3702.283	3409.56	3409.56	3542.616	3542.616
C880	2514.758	2468.189	2468.189	2514.758	2514.758
C1355	3506.026	3296.462	3296.462	3502.699	3502.699

If the chip is targeted for area, from the above results it is clear that multi threshold technique targeting for zero leakage power without clock gating is suggestible. Clock gating technique involves extra hardware and consequently leads to occupy more area. If the chip is delay closure, the technique which involves clock gating gives good results due to suppression of unwanted pulses by optimizing the architecture.

Table 3: Performance Comparison for Delay (ps) Design Closure for Existing Techniques.

Ckt	GS&B	CG	MV _t
C432	371	592	565
C499	527	291	647
C880	370	316	671
C1355	401	458	611

Table 4: Performance Comparison for Delay (ps) Design Closure for Proposed Techniques.

Ckt	MV _t +CG	MV _t -Zero Leakage		MV _t -Zero Dynamic	
		w/o CG	with CG	w/o CG	with CG
C432	597	454	494	565	597
C499	312	625	625	655	655
C880	671	672	672	671	671
C1355	611	583	583	623	623

Table 5: Performance Comparison for Power (mW) Design Closure for Existing Techniques.

Ckt	GS&B	CG	MVt
C432	178.770	98.134	98.218
C499	760.545	726.129	716.178
C880	387.432	292.582	270.130
C1355	902.971	767.997	669.225

Table 6: Performance Comparison for Power (mW) Design Closure for Proposed Techniques.

Ckt	MVt+CG	MVt-Zero Leakage		MVt-Zero Dynamic	
		w/o CG	with CG	w/o CG	with CG
C432	96.897	107.666	104.238	98.218	96.897
C499	801.560	758.454	758.454	721.952	721.952
C880	270.067	279.861	279.852	270.130	270.067
C1355	669.225	693.970	693.970	668.509	668.509

Table 7: Performance Comparison for % of Glitch Power Reduction for Existing Techniques.

Ckt	GS and B	CG	MVt
C432	6.07	18.76	23.19
C499	4.30	8.47	2.31
C880	5.81	12.28	4.10
C1355	3.01	9.19	1.78

Table 8: Performance Comparison for % of Glitch Power Reduction for Proposed Techniques.

Ckt	MVt+CG	MVt-Zero Leakage		MVt-Zero Dynamic	
		w/o CG	with CG	w/o CG	with CG
C432	20.80	9.27	14.42	23.19	20.80
C499	2.31	2.62	2.98	3.62	4.70
C880	3.96	1.30	7.07	4.10	13.67
C1355	1.84	2.38	3.13	3.84	10.11

If the chip's power is the main design objective, multi threshold targeting zero dynamic with clock gating is suggestible. By targeting the chip for zero dynamic, cells from existing three libraries typical, fast and slow are appropriately chosen by suppressing switching activity. Having considered the pros and cons of all the above techniques we can finally say, for multi objective design closure against glitches, multi threshold with clock gating technique is suggestible.

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